Homework 1

Objectives: training the R script typing; better understand how to calculate the RMSE for training and testing, and comparing them.

Type and study the scripts bellow. At the end you will find your homework activities. Have fun!!

```
LoadLibraries=function (){
  library (ISLR)
  library (MASS)
  print ("The libraries have been loaded.")
  }

LoadLibraries()

saving_r2 <- function(lm_output){
  obj1 <- summary(lm_output)
  r2 <- obj1$r.squared
  return(r2)
}

saving_r2(lm.fit2)

my_rmse <- function(original_y, yhat){
  rmse <- sqrt(mean((original_y-yhat)^2))
  return(rmse)
}

# testing

yy <- c(1, 2, 3, 4)
yyhat <- c(1, 1, 2, 3)

my_rmse(yy, yyhat)

#############################
# upload the Boston dataset
#############################

dd <- Boston # renaming the Boston dataset to easy notation below
dim(dd)

### fitting a simple linear model with n = 506

fit1 <- lm(medv ~ lstat, data = dd)
fit1

# finding the rmse for this model:

my_rmse(dd$medv, fit1$fitted.values)


### Quadratic Model

fit2 <- lm(medv~lstat + I(lstat^2), data = dd)
my_rmse(dd$medv, fit2$fitted.values)
```

```
### Cubic Model

fit3 <- lm(medv~lstat+I(lstat^2)+I(lstat^3), data = dd)
my_rmse(dd$medv, fit3$fitted.values)

# as we can see, the RMSE decreases as we consider a more flexible model
# try doing this for other polinomials

#----------------------------------------------------------------------------
# now, our main interest is better understanding the idea of RMSE on a training
# dataset and how it compares with the RMSE calculated on a testing dataset.
# for this, let´s run the following toy example:

#######################################
# trying the validation set approach
#######################################

# first we select which rows of our dataset will belong to the training population

train <- sample(dim(dd)[1], size = dim(dd)[1]/2) # 50% training; 50% testing

# now, we split the dataset (dd) in training and testing

dd_train <- dd[train, ] # select all the rows that match the train object
dim(dd_train)

# now, fit each model on this training dataset. Obtain the RMSE for training and testing

#####################################################
### Starting with a simple linear regression model
#####################################################

fit1 <- lm(medv ~ lstat, data = dd_train)
fit1

# RMSE for training

my_rmse(dd$medv[train], fit1$fitted.values) # RMSE_train - grau 1

# calculating RMSE on the testing dataset

yhat <- fit1$coefficients[1] + fit1$coefficients[2] * dd$lstat[-train]

# alternatively, we could have used the predict function in R to obtain this yhat:

yhat_other <- predict(fit1, newdata = dd[-train,])

# comparing both vectors of predicted y
```

```
head(cbind(yhat, yhat_other)) # showing the first 7 entries. The
values are the same

my_rmse(dd$medv[-train], yhat) # RMSE_test - grau 1

plot(dd$medv[-train], yhat)
abline(0,1)

################################
# now for the quadratic model
################################

fit2 <- lm(medv ~ lstat + I(lstat^2), data = dd_train)
fit2

my_rmse(dd$medv[train], fit2$fitted.values) # RMSE_train - grau 2

# calculating RMSE on the testing dataset

yhat2 <- fit2$coefficients[1] + fit2$coefficients[2] * dd$lstat[-
train] +
  fit2$coefficients[3] * (dd$lstat^2)[-train]

# or:

yhat2_other <- predict(fit2, dd[-train,])
head(cbind(yhat2, yhat2_other)) # showing the first 7 entries. The
values are the same

my_rmse(dd$medv[-train], yhat2) # RMSE_test - grau 2

plot(dd$medv[-train], yhat2)
abline(0,1)

# now that we know the predict function can be used to predict y
values from a fitted
# model, we can use it on the rest of this script

############################
# now for the cubic model
############################

fit3 <- lm(medv ~ lstat + I(lstat^2) + I(lstat^3), data = dd_train)
fit3

my_rmse(dd$medv[train], fit3$fitted.values) # RMSE_train - grau 3

# calculating RMSE on the testing dataset
yhat3 <- predict(fit3, dd[-train,])

my_rmse(dd$medv[-train], yhat3) # RMSE_test - grau 3

plot(dd$medv[-train], yhat3)
abline(0,1)

################################
# Now, your homework activity:
################################

# 1) keep doing the calculation of RMSE_train and RMSE_test for the
other polinomial
```

```
    # models (degree 4, 5, 6, 7, 8, 9, and 10) on this same training
and test dataset

    # save the values RMSE_train in a vector named vector_RMSE_train;
save the values
    # RMSE_test in a vector named vector_RMSE_test

    # 2) make a plot where the x-axis is the polinomial degree, and the
y-axis is the
    # vector_RMSE_train and vector_RMSE_test (on the same plot)
    # obs: I do expect you will have a plot very similar to the one
shown in
    # FIGURE 2.9 of our textbook.

    # 3) comment your results
```

Delivery deadline: the week following the holiday. You will have to respond to an electronic file (E.g., R script) and return it to peternelli@ufv.br. Your filename should be stated as follows: WH1_your_first_name.docx. The same filename should be the subject of your email to me, with the file attached.